

# **BECE102L – Digital Systems Design**

## **Module 4 – Design of Data Path Circuits**

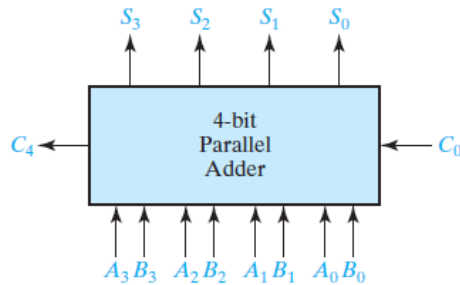
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*N-bit Parallel Adder/Subtractor, Carry Look Ahead Adder, Unsigned Array Multiplier, Booth Multiplier, 4-Bit Magnitude comparator. Modeling of data path circuits using Verilog HDL.*

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# Design of 4-bit parallel binary adder

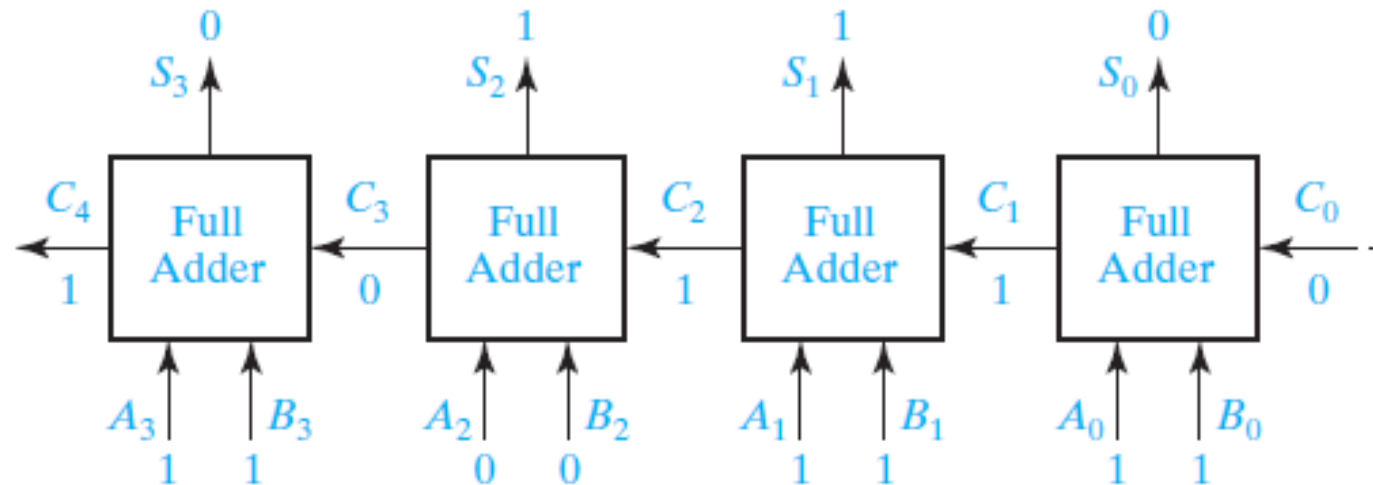
- A parallel adder that adds two N-bit unsigned binary numbers and a carry input to give a N-bit sum and a carry output.
- If  $N = 4$ , then it is 4-bit parallel binary adder.



- One approach would be to construct a truth table with nine inputs and five outputs and then derive and simplify the five output equations.
- But, each equation would be a function of nine variables before simplification and this approach would be very difficult, and the resulting logic circuit would be very complex.

# Design of 4-bit parallel binary adder

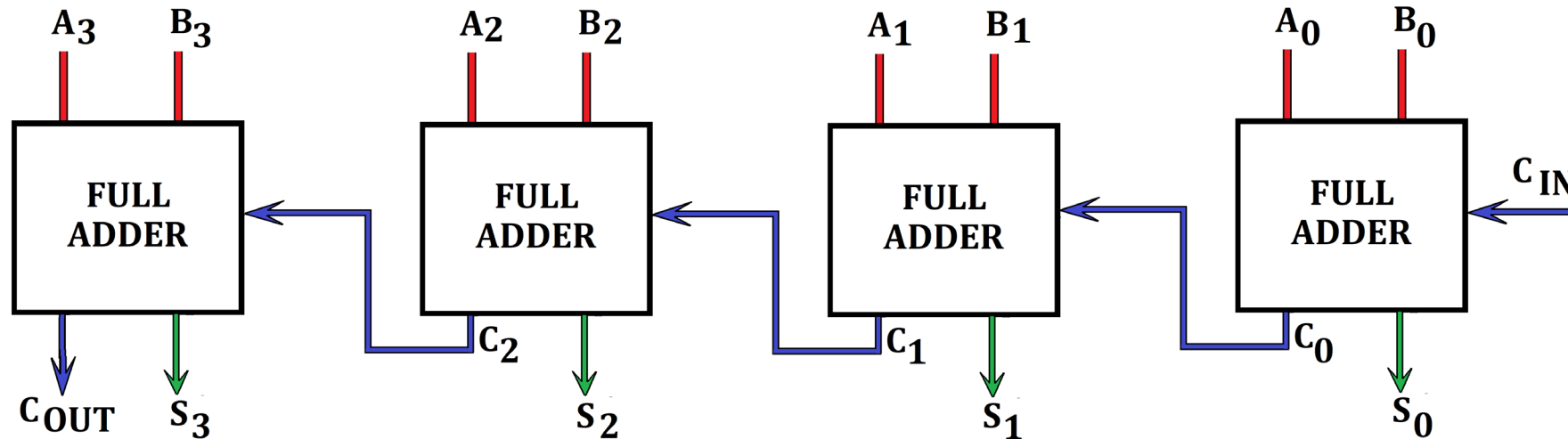
- A better method is to design a logic module that adds two bits and a carry, and then connect four of these modules together to form a 4-bit adder.



- If the number of bits is large ( $N$  is large), a parallel binary adder of the type may be rather slow because the carry generated in the first cell might have to propagate all of the way to the last cell.
- The number of gate levels for the carry propagation can be found from the circuit of the full adder.

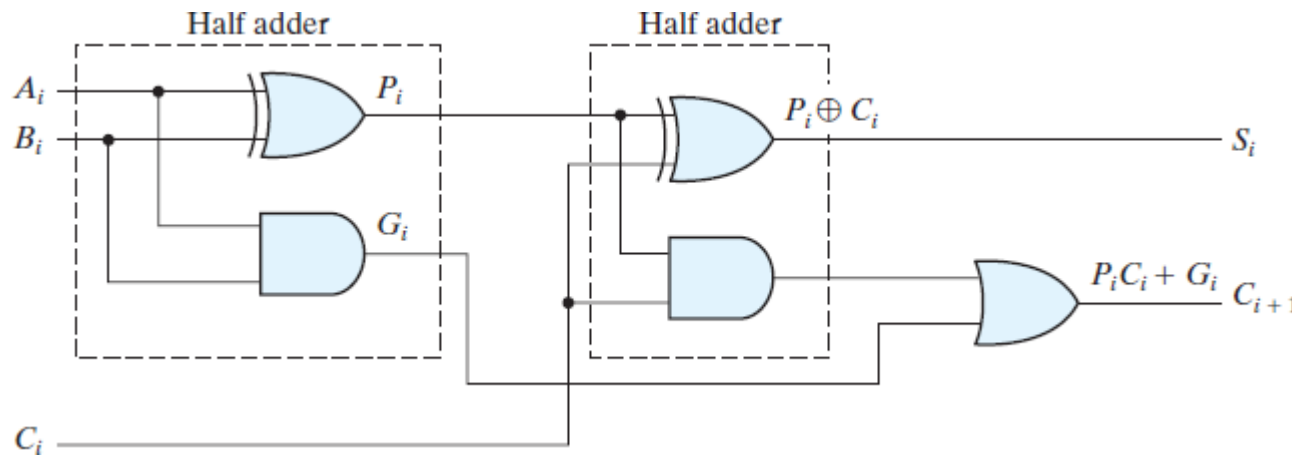
# 4-bit parallel or RIPPLE CARRY ADDER binary adder

|           |       |       |       |          |
|-----------|-------|-------|-------|----------|
|           | $A_3$ | $A_2$ | $A_1$ | $A_0$    |
|           | $B_3$ | $B_2$ | $B_1$ | $B_0$    |
|           | $C_2$ | $C_1$ | $C_0$ | $C_{IN}$ |
| $C_{OUT}$ | $S_3$ | $S_2$ | $S_1$ | $S_0$    |



# Carry Lookahead

- A parallel adder that adds two N-bit unsigned binary numbers and a carry input to give a N-bit sum and a carry output.



- The signal from the input carry  $C_i$  to the output carry  $C_{i+1}$  propagates through an AND gate and an OR gate, which constitute two gate levels.
- If there are four full adders in the adder, the output carry  $C_4$  would have  $2 * 4 = 8$  gate levels from  $C_0$  to  $C_4$ .
- For an  $n$ -bit adder, there are  $2n$  gate levels for the carry to propagate from input to output.

# Cont...

| A | B | C <sub>in</sub> | C <sub>out</sub> |
|---|---|-----------------|------------------|
| 0 | 0 | 0               | 0                |
| 0 | 0 | 1               | 0                |
| 0 | 1 | 0               | 0                |
| 0 | 1 | 1               | 1                |
| 1 | 0 | 0               | 0                |
| 1 | 0 | 1               | 1                |
| 1 | 1 | 0               | 1                |
| 1 | 1 | 1               | 1                |

Either A or B is '1', but  
C<sub>in</sub> is always 1

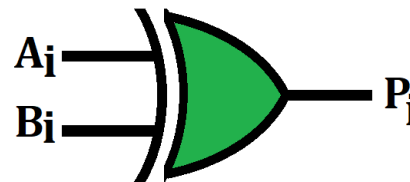
When both A and B are '1',  
then we have carried  
irrespective of any C<sub>in</sub>.

$$C_i = [AB] + [(A \otimes B)] C_{i-1}$$

Carry Generate (G<sub>i</sub>)



Carry Propagate (P<sub>i</sub>)



# Cont...

$$C_i = A_i B_i + (A_i \otimes B_i) C_{i-1}$$

$$C_i = G_i + P_i C_{i-1}$$

$$C_0 = G_0 + P_0 C_{in}$$

$$C_1 = G_1 + P_1 C_0 = G_1 + P_1 (G_0 + P_0 C_{in})$$

$$C_1 = G_1 + G_0 P_1 + P_1 P_0 C_{in}$$

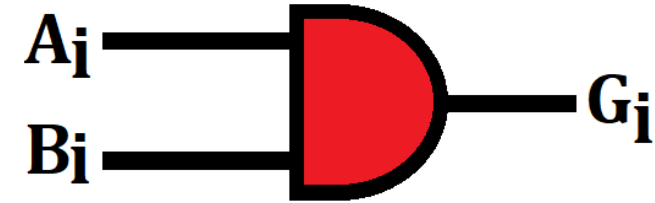
$$C_2 = G_2 + P_2 (G_1 + G_0 P_1 + P_1 P_0 C_{in})$$

$$C_2 = G_2 + P_2 G_1 + G_0 P_1 P_2 + P_0 P_1 P_2 C_{in}$$

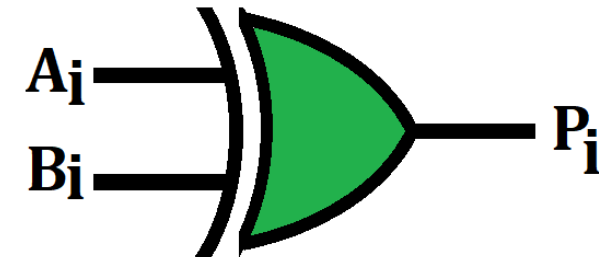
$$C_3 = G_3 + P_3 (G_2 + P_2 G_1 + G_0 P_1 P_2 + P_0 P_1 P_2 C_{in})$$

$$C_3 = G_3 + P_3 G_2 + P_3 P_2 G_1 + G_0 P_1 P_2 P_3 + P_0 P_1 P_2 P_3 C_{in}$$

Carry Generate ( $G_i$ )



Carry Propagate ( $P_i$ )



# Carry Lookahead

- There are several techniques for reducing the carry propagation time in a parallel adder. The most widely used technique employs the principle of carry lookahead logic

$$P_i = A_i \oplus B_i$$

$$G_i = A_i B_i$$

$$S_i = P_i \oplus C_i$$

$$C_{i+1} = G_i + P_i C_i$$

- $G_i$  is called a carry generate, and it produces a carry of 1 when both  $A_i$  and  $B_i$  are 1, regardless of the input carry  $C_i$ .
- $P_i$  is called a carry propagate, because it determines whether a carry into stage  $i$  will propagate into stage  $i + 1$ .



# Carry Lookahead

- There are several techniques for reducing the carry propagation time in a parallel adder. The most widely used technique employs the principle of carry lookahead logic

$$C_0 = \text{input carry}$$

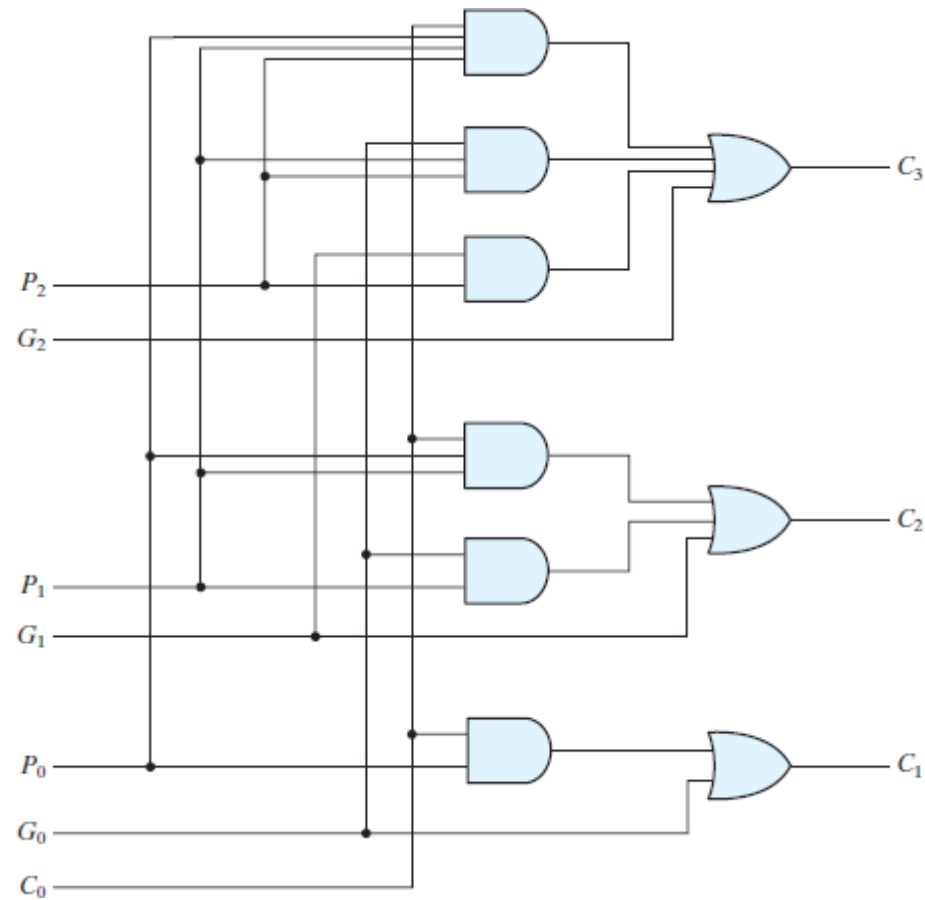
$$C_1 = G_0 + P_0C_0$$

$$C_2 = G_1 + P_1C_1 = G_1 + P_1(G_0 + P_0C_0) = G_1 + P_1G_0 + P_1P_0C_0$$

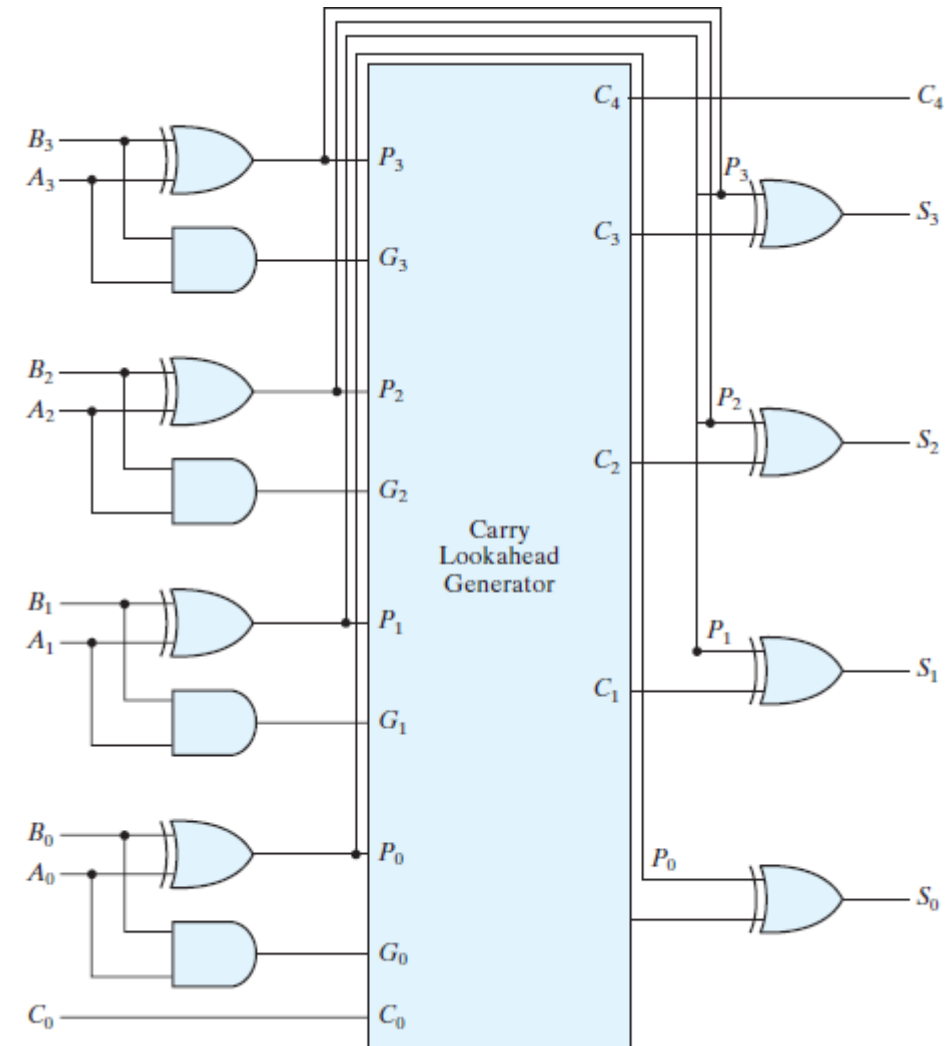
$$C_3 = G_2 + P_2C_2 = G_2 + P_2G_1 + P_2P_1G_0 = P_2P_1P_0C_0$$

- The three Boolean functions for  $C_1$ ,  $C_2$ , and  $C_3$  are implemented in the carry lookahead generator
- All output carries are generated after a delay through two levels of gates.
- Thus, outputs  $S_1$  through  $S_3$  have equal propagation delay times.

# Carry Lookahead Adder



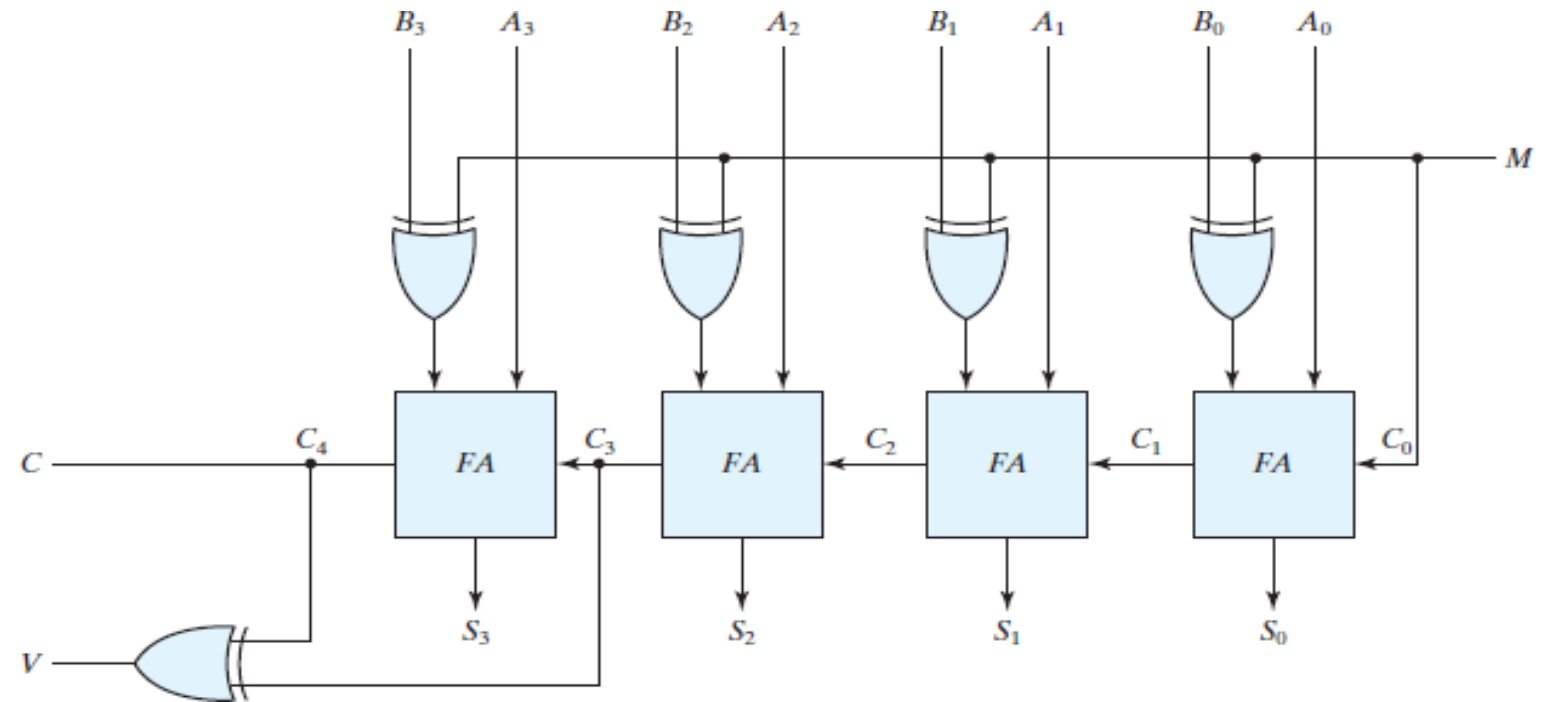
Logic diagram of carry lookahead generator



Four-bit adder with carry lookahead

# Four-bit adder-subtractor

- The circuit for subtracting  $A - B$  consists of an adder with inverters placed between each data input  $B$  and the corresponding input of the full adder.
- The input carry  $C_0$  must be equal to 1 when subtraction is performed.



**When  $M = 0$ , the circuit is an adder, and when  $M = 1$ , the circuit becomes a subtractor.**

The operation thus performed becomes  $A$ , plus the 1's complement of  $B$ , plus 1. This is equal to  $A$  plus the 2's complement of  $B$ .

For unsigned numbers, that gives  $A - B$  if  $A \geq B$  or the 2's complement of  $(B - A)$  if  $A < B$ . For signed numbers, the result is  $A - B$ , provided that there is no overflow.

# Overflow

When two numbers with  $n$  digits each are added and the sum is a number occupying  $n + 1$  digits, we say that an overflow occurred.

carries:

+70

+80

+150

0 1

0 1000110

0 1010000

1 0010110

carries:

−70

−80

−150

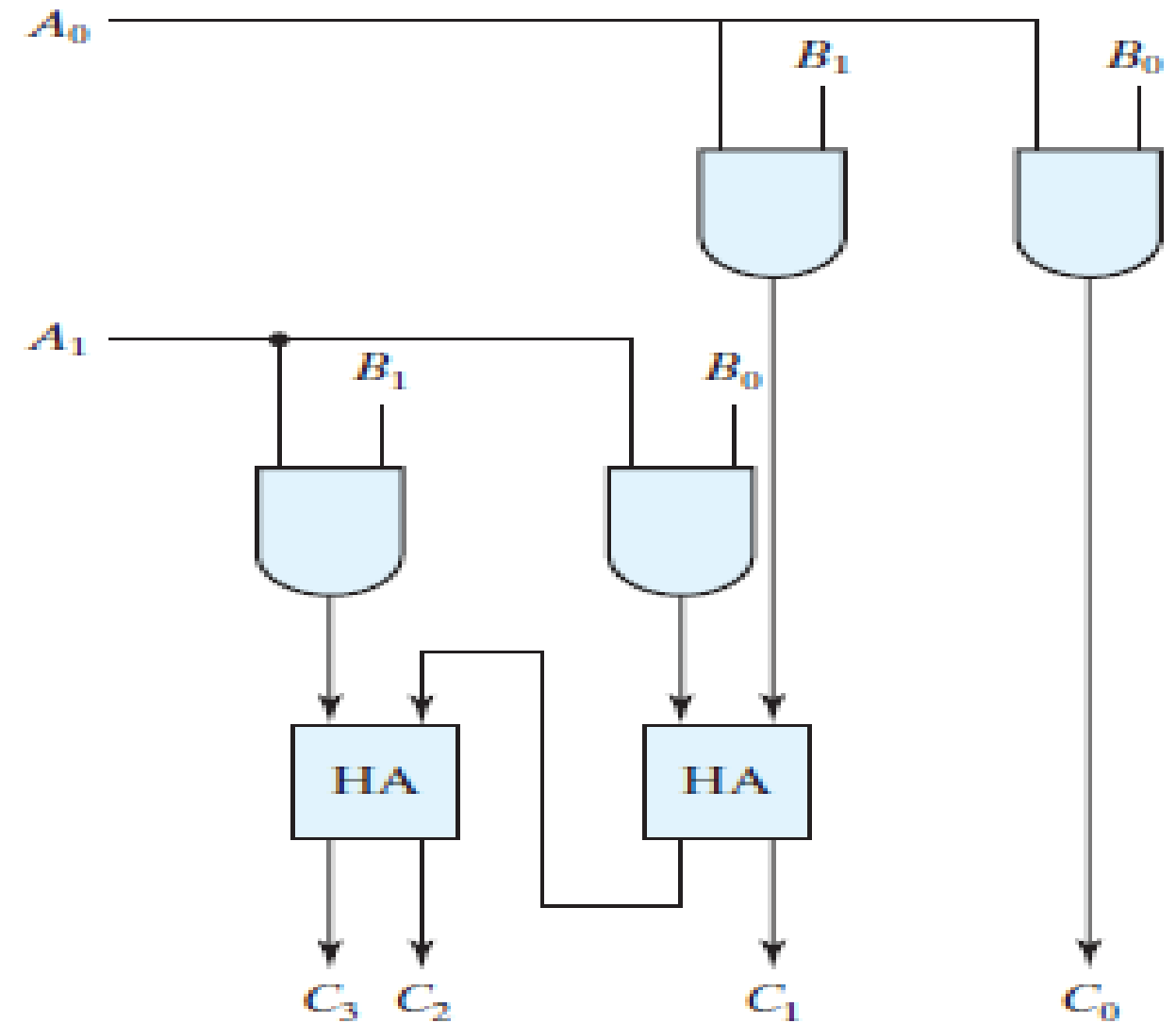
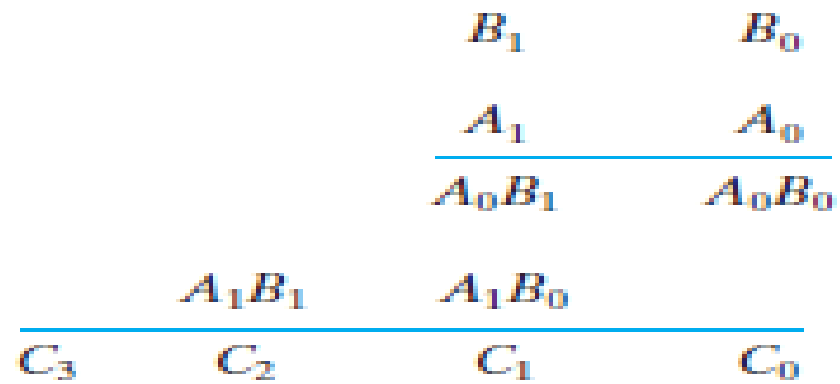
1 0

1 0111010

1 0110000

0 1101010

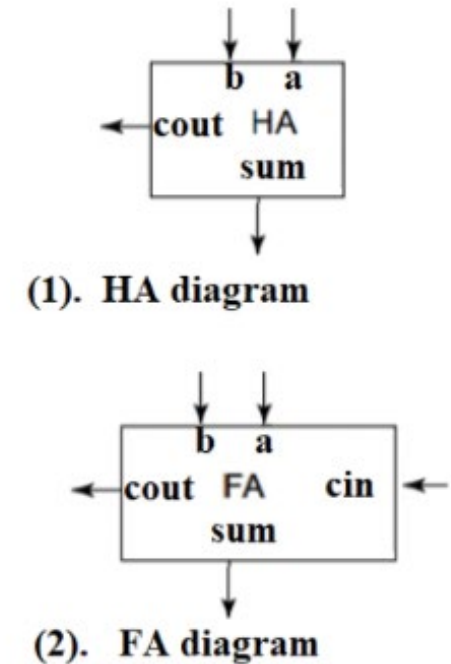
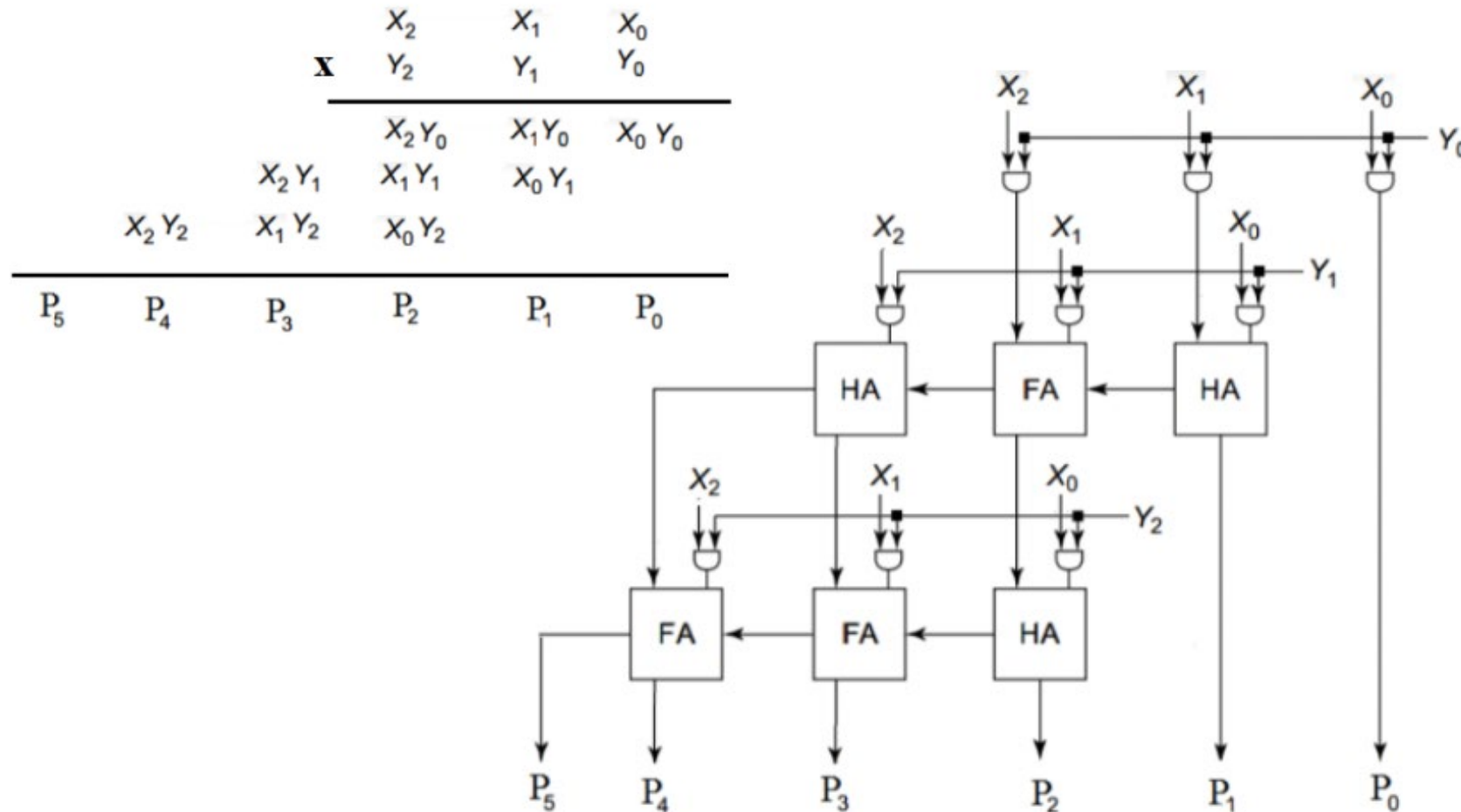
# 2x2 Binary Multiplier



# 3x3 Binary Multiplier

## Part 1: 3 By 3 Binary Combinational Array Multiplier

The binary combinational multiplier diagram:

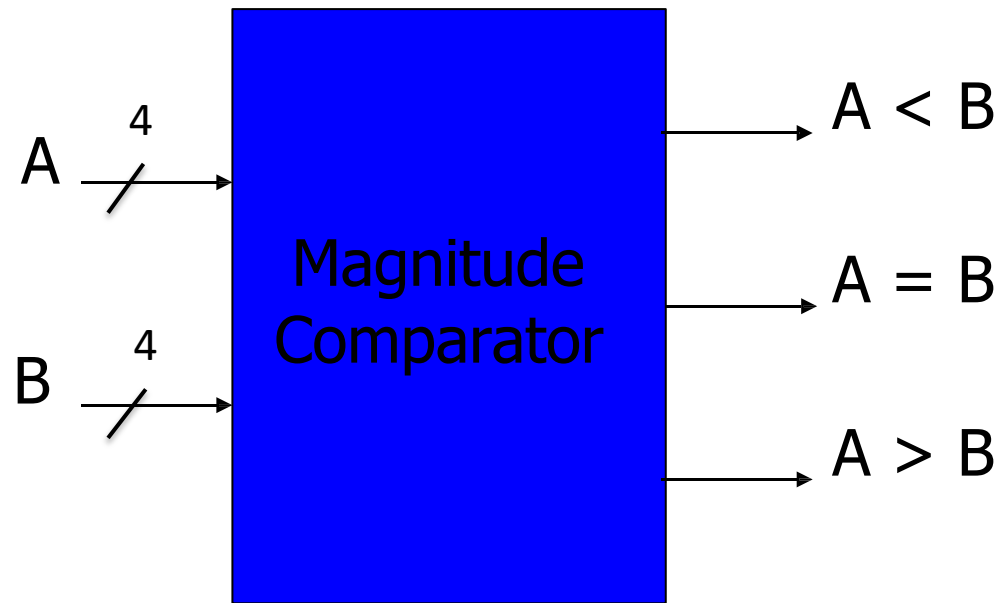


# Magnitude Comparator

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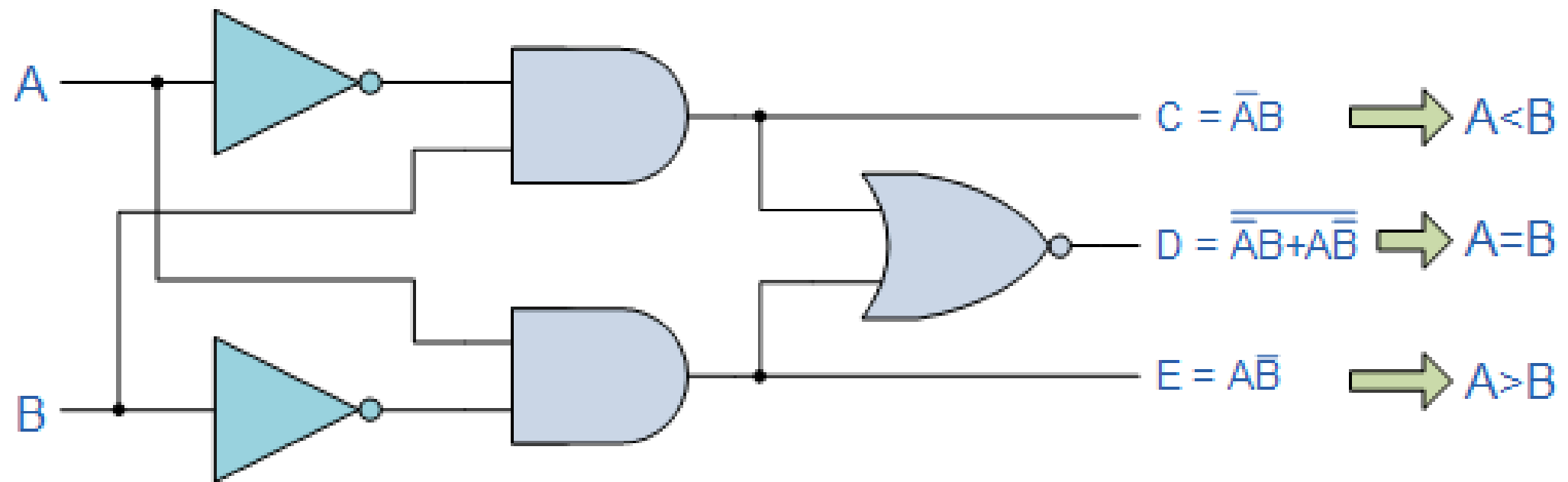
- Magnitude comparators compare two multi-bit binary numbers
- Magnitude comparators allow for data comparison and can be built using and-or gates
- Greater/less than function requires more hardware than equality
- The comparison of two numbers gives the outputs:  $A > B$ ,  $A = B$ ,  $A < B$

# Magnitude Comparator

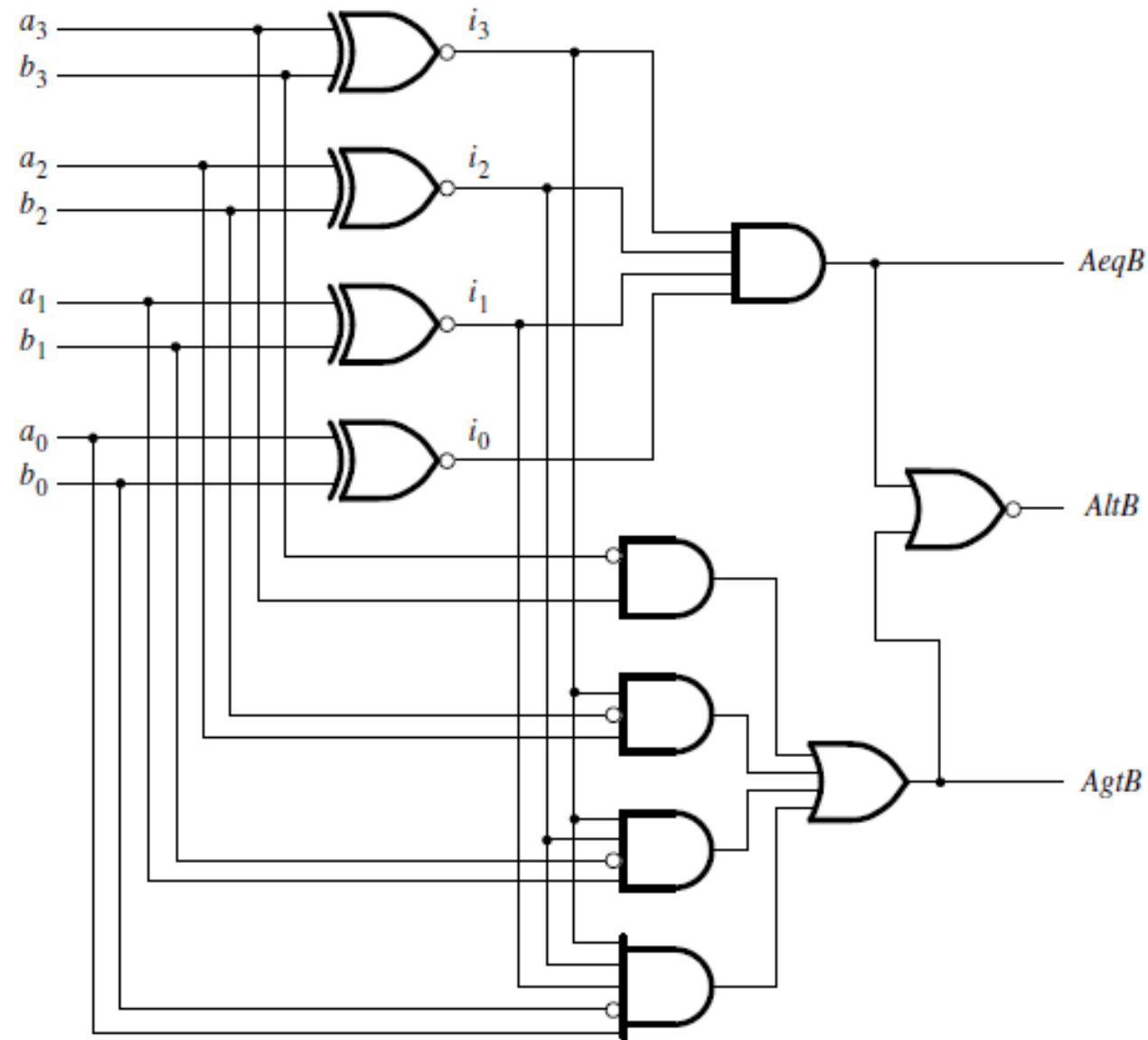


| Inputs |   | Outputs |         |         |
|--------|---|---------|---------|---------|
| B      | A | $A > B$ | $A = B$ | $A < B$ |
| 0      | 0 | 0       | 1       | 0       |
| 0      | 1 | 1       | 0       | 0       |
| 1      | 0 | 0       | 0       | 1       |
| 1      | 1 | 0       | 1       | 0       |





# Logic Circuit for 4 bit Magnitude Comparison



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